

1. Executive Summary

This document serves as the final reference guide for the Healthcare SNF (Skilled Nursing Facility) data pipeline. It traces the lifecycle from the initial business requirements to the final Star Schema delivered in the Gold layer, documenting which metrics were successfully tackled and which were omitted due to data constraints.

2. Final Business Requirements Mapping

The project originally defined 4 core business questions. Because the source dataset (CMS Nursing Home Compare) focuses on compliance and quality rather than internal hospital operations (like payroll or shift-by-shift tracking), several questions were "reframed" during the Silver layer architecture phase.

✓ Question 1: Relationship between nurse staffing levels and occupancy rates

- **Status:** Answered
- **Gold Table Used:** FCT_STAFFING_OCCUPANCY joined with DIM_PROVIDERS
- **Columns Used:**
 - OCCUPANCY_RATE : Built from raw avg_daily_residents / certified_bed_count , capped at 110% to prevent denominator overflow errors.
 - REPORTED_TOTAL_NURSE_HOURS_PRD : Maps the daily workload per resident.
- **Why:** Combining these columns lets us build scatter plots to definitively show if high occupancy leads to squeezed/low staffing hours.

✓ Question 2: Which facilities have the highest overtime hours?

- **Status:** Reframed and Answered
- **Reframing:** Overtime data does not exist in standard CMS datasets. Reframed to investigate **Burnout and Turnover**.
- **Gold Table Used:** FCT_TURNOVER_ANALYSIS joined with DIM_PROVIDERS
- **Columns Used:**
 - TOTAL_NURSE_TURNOVER_PCT and RN_TURNOVER_PCT : Maps the percentage of nursing staff that left the facility.
 - REPORTED_TOTAL_NURSE_HOURS_PRD : Sourced from the providers table to see if low staffing correlates with high turnover.
- **Why:** Turnover is the most direct indicator of workforce hostility or "burnout," providing an even stronger operational insight than plain overtime.

✓ Question 3: Average staffing levels by state and hospital type?

- **Status:** Answered
- **Gold Table Used:** FCT_STAFFING_OCCUPANCY interacting with DIM_PROVIDERS
- **Columns Used:**
 - STAFFING_TO_NATIONAL_AVG_RATIO and STAFFING_TO_STATE_AVG_RATIO : Derived metrics connecting facility performance to state/national benchmarks.
 - STATE_CODE and OWNERSHIP_TYPE (from DIM_PROVIDERS): Used as the primary grouping dimensions in all dashboards.
- **Why:** The STAFFING_TO_NATIONAL_AVG_RATIO acts as a powerful normalized index (e.g., 0.85x), making it incredibly simple for BI tools to highlight understaffed facilities without confusing absolute numbers.

✓ Question 4: Trends in patient length of stay?

- **Status:** Reframed and Answered
- **Reframing:** No patient-level admission/discharge date data exists. Reframed to investigate **Readmission Rates and Quality**.
- **Gold Table Used:** FCT_QUALITY_READMISSION
- **Columns Used:**
 - PERFORMANCE_READMISSION_RATE : Measures the 30-day readmission risk based on Value-Based Purchasing (VBP) scores.
 - REPORTED_TOTAL_NURSE_HOURS_PRD (brought over from providers): Mapped to see if high staffing prevents hospital readmissions.
- **Why:** Readmission rates are the universal Medicare standard for care quality and length-of-stay proxy metrics.

3. Final Metrics Mapping

The original project scope identified several core metrics to measure facility performance. Here is how the 5 finalized metrics were technically implemented in the Gold Layer schemas to serve the BI Dashboard.

✓ Metric 1.1: Nurse-to-Patient Ratio

- **Status:** Answered
- **Gold Table Used:** FCT_STAFFING_OCCUPANCY
- **Columns Used:** REPORTED_TOTAL_NURSE_HOURS_PRD , STATE_AVG_TOTAL_NURSE_HOURS_PRD
- **Why:** Rather than a simple ratio, this normalizes the staffing level into "hours per resident per day" and arrays it against state benchmarks to measure standard of care.

✓ Metric 2.3: Staffing vs. Occupancy

- **Status:** Answered
- **Gold Table Used:** `FCT_STAFFING_OCCUPANCY`
- **Columns Used:** `REPORTED_TOTAL_NURSE_HOURS_PRD` , `OCCUPANCY_RATE`
- **Why:** Answers how close facilities are to their physical limits and how that stress interacts with the size of their nursing staff.

✓ **Metric 2.5: Lowest Staffed Facilities**

- **Status:** Answered
- **Gold Table Used:** `FCT_STAFFING_OCCUPANCY`
- **Columns Used:** `STAFFING_TO_NATIONAL_AVG_RATIO`
- **Why:** Creates an easy filter to spot "outliers"—any facility with a ratio mathematically approaching 0 is severely under national staffing thresholds.

✓ **Metric 3.3: Readmission Rates (Quality of Care)**

- **Status:** Answered
- **Gold Table Used:** `FCT_QUALITY_READMISSION`
- **Columns Used:** `PERFORMANCE_READMISSION_RATE` , `READMISSION_TO_NATIONAL_AVG_RATIO`
- **Why:** Utilizes the CMS Value-Based Purchasing (VBP) score to evaluate the direct risk of a facility sending a resident back to the hospital.

✓ **Metric 3.5: Correlation between Staffing Levels and Readmission**

- **Status:** Answered
- **Gold Table Used:** `FCT_QUALITY_READMISSION`
- **Columns Used:** `REPORTED_TOTAL_NURSE_HOURS_PRD` , `PERFORMANCE_READMISSION_RATE`
- **Why:** Joins demographic staffing logic with health performance logic to create our "Quality Impact" scatter plot visualization.

4. Architectural Summary

The final architecture fully adheres to the Medallion Data Engineering paradigm:

- **Bronze:** All tables re-ingested strictly as `VARCHAR` to prevent `INFER_SCHEMA` logic from breaking on bad string artifacts (e.g., '---').
- **Silver:** Safe casting logic (`TRY_TO_DECIMAL` , `TRY_TO_NUMBER` , etc.), null-handling, column pruning, and deduplication handled cleanly into 3 pristine core models.
- **Gold:** Shifted to a normalized **Star Schema** (Fact/Dimension structure).
 - Models are materialized as heavily-optimized `Views` to ensure **ZERO incremental storage cost** for Snowflake—a highly cost-effective pattern for this analytical profile.

- Implemented `INNER JOIN` referential integrity checks (to block 170 orphaned VBP records seamlessly).
- Deployed `LEAST(1.1, ...)` capping logic to handle dirty denominator capacities gracefully.